

- 16 The table below shows the number of cars and motorcycles passing through an Electronic Road Pricing (ERP) gantry on certain days of the week from 7.30 am to 7.55 am.

	Cars	Motorcycles
Wednesday	320	120
Thursday	380	100
Friday	410	130
Charges per vehicle	\$2	\$0.50

- (a) Represent the number of vehicles passing through the ERP gantry in a 3×2 matrix $\mathbf{V}$.

$$\text{Answer } \mathbf{V} = \begin{pmatrix} & & \end{pmatrix} \quad [1]$$

- (b) Write down a matrix $\mathbf{C}$ such that the product $\mathbf{P} = \mathbf{VC}$ represents the total charges incurred by all vehicles on each of the days.

$$\text{Answer } \mathbf{C} = \quad [1]$$

- (c) Evaluate the matrix $\mathbf{P} = \mathbf{VC}$.

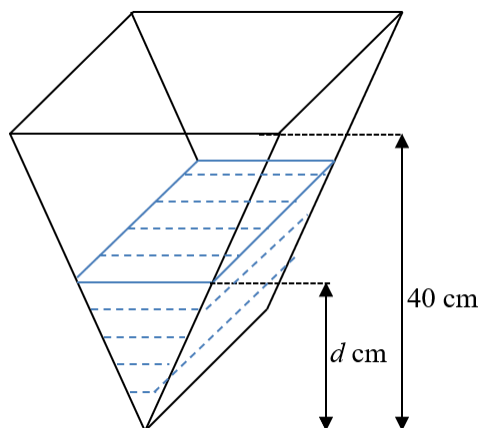
$$\text{Answer } \mathbf{P} = \quad [1]$$

- (d) $\mathbf{A}$ is a matrix such that $\mathbf{A} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \mathbf{P}$.
State what $\mathbf{A}$ represents.

Answer _____

[1]

17



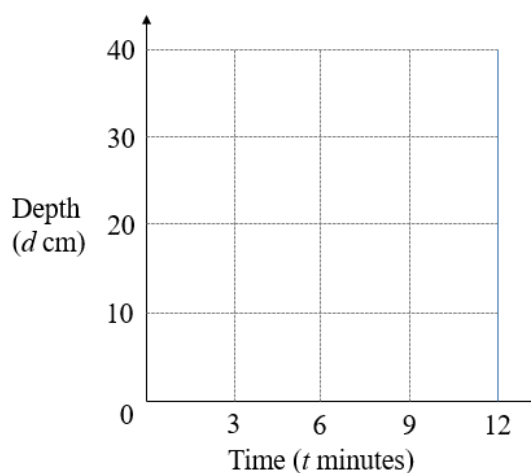
The diagram shows a container in the shape of a prism with a triangular cross-section. The container has a height of 40 cm. Water is poured into the empty container at a constant rate. It takes 12 minutes to fill the container completely. After t minutes, the depth of the water is d cm.

- (a) Find the value of t when $d = 20$.

Answer $t =$ _____ [2]

- (b) On the grid, sketch the graph showing how the depth varies during the 12 minutes.

Answer



[2]

